Exercice 4 :

. Under white noise assumption of $\{r_k\}$, we have

$$\sigma_r = SE(r_k) = \frac{1}{\sqrt{121}} = 0,09$$

using the $\pm 2\sigma_r$ 95% confidence interval yields that $\hat{\phi}_{33} = 0,08 < 2 \times 0,09 = 0,18$ but $\hat{\phi}_{11}$ and $\hat{\phi}_{22}$ remain significative.

Conclusion the time serie is an ARIMA(2,0,0) = AR(2)